

Do Politicians' Words Have a Dollar Value?

Does daily political climate sentiment influence the Greenium in India & Indonesia?

- Climate speeches → sovereign green bond yields in India & Indonesia
- Gap: existing literature uses monthly data; politics moves daily
- Answer: signaling noise costs money; raw sentiment does not

When Finance Ministry says "green" but Energy Ministry says "coal"

- Green bonds = sovereign loans pledged to green projects
- Greenium = yield discount investors accept for the green label
- India, Indonesia: large coal economies issuing green bonds — structural paradox

Credibility Premium
*Consistent signals →
lower borrowing costs*

Credibility Gap
*Contradictory signals →
trust penalty*

1 bp saving on \$1bn bond = \$1m/yr less interest
Emerging markets need every bp they can get

Daily bond yields + Daily political sentiment, 2020–2024

1,455

Panel observations

4

Green–Conventional twin pairs

120,000

GDELT Articles

- Bond yields: Refinitiv — daily YTM, green vs. conventional twin
- Sentiment: GDELT + ClimateBERT — daily political climate tone
- Controls: VIX, CDS spreads, Exchange Rates

Twin-bond DiD: green bond vs. maturity-matched conventional twin

- Same issuer, same currency, same maturity → default & currency risk cancel out
- Greenium = $YTM_{\text{green}} - YTM_{\text{conventional}}$ is the outcome
- Staggered DiD: *Events*
 - India Jan 2023 (green bond auction);
 - Indonesia Nov 2022 (JETP/G20)
- Event study (± 10 weeks) validates parallel trends + tests pre-announcement drift

All regressors first-differenced and Standardized
Standard errors are HC3 heteroskedasticity-robust throughout
Signaling noise = std of daily article scores

Noise is priced; sentiment alone is not

H1 — Positive sentiment → wider greenium

$\beta = +0.0009$, not statistically significant

✗ null — sentiment doesn't move
Greenium

*H2 — Higher signaling noise → narrower
greenium*

$\beta = -0.0031^{*}$**

✓ significant at 1%

*H3 — Policy events → structural shift in
greenium*

Post = -0.0301 (JETP)

✓ Indonesia detected; India null

Parallel trends

0/9 pre-event Statistically significant

✓ DiD identification valid

Markets don't price words — they price consistency

- Higher noise (MoF vs. MoE contradiction) widens Greenium spread — markets demand consistency premium, not just green rhetoric
- Indonesia's JETP shock detectable; India's green bond auction null → international validation matters more than domestic auction
- Null sentiment result supports weak-form efficiency for climate discourse
- ClimateBERT robustness limited (N=20, 1.4% coverage) — GDELT remains primary measure

Next Goal is to Reduce the Limitations

GDELT is not green-bond specific

ClimateBERT coverage too sparse

Small sample — 2 countries, 4 bond pairs

Secondary market only

Political credibility is a fiscal variable

What need does this fill?

First high-frequency causal evidence on Greenium dynamics in Asian emerging markets

What did I do?

DiD + event study with NLP sentiment on daily green vs. conventional bond pairs (India, Indonesia)

Main takeaway

Signaling noise — not sentiment — costs money; ~3bp per 1-SD noise increase

Full replication available on GitHub



Raw Data · Processed sentiment CSV · Panel Data · Notebook · Plots · Chapters

References

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Data sources: Refinitiv Workspace (bond yields) · GDELT BigQuery (sentiment) · FRED (VIX) · Investing.com (CDS spreads)

Bond Inventory — All 8 Instruments

Pair	Green Bond (ISIN)	Conventional Twin	Currency	Mean Greenium
India 10Y	IN0020220144 (Jan 2023)	IN0020190362 (Jan 2020)	INR	+0.138 pp
India 5Y	IN0020230143 (Nov 2023)	IN072629G (Jan 2020)	INR	−0.000 pp
Indonesia 10Y	US71567RAV87 (Jun 2022)	US455780DN36 (Sep 2022)	USD	−0.055 pp
Indonesia 2033	US71567RAY27 (Nov 2023)	US455780DR40 (Jan 2023)	USD	+0.055 pp

Source: Refinitiv Workspace via CEU institutional access

Sentiment Pipeline & ClimateBERT Coverage

GDELT V2Tone (primary)

- 1 Query GDELT BigQuery — India/Indonesia × climate themes
- 2 Sample 1,000 articles/country/month
- 3 Parse V2Tone → normalise to [−1, +1]
- 4 Aggregate: daily mean (sentiment) + daily std (signaling noise)

3654	120,000	India 0.33 · Indonesia 0.40
Country-days	Total Articles	Mean noise σ

ClimateBERT (robustness only)

Model:	distilroberta-base-climate-sentiment
Stage 1:	Climate detector → filter headlines
Stage 2:	opportunity (+1) / neutral / risk (−1)
Coverage:	52 matched days (1.4% of panel)
After FD:	Only 20 obs → sparse
India r:	r = +0.276 vs V2Tone (weak positive)
Indonesia r:	r = −0.037 vs V2Tone (diverges)
Verdict:	Directionally consistent for India; inconclusive overall

Full Regression Table

Variable	M1	M2	M3	M4	M5
Δ Sentiment (H1)	0.0008	0.0009	0.0005	0.0005	0.0009
Δ Signaling Noise (H2)	—	—	−0.0031***	−0.0030***	—
Post Treatment (H3)	—	—	—	—	−0.0298
R^2	0.0004	0.0085	0.0129	0.0129	0.02
Adj. R^2	−0.0024	0.0037	0.0074	0.0067	−0.0042

* p<0.1 ** p<0.05 *** p<0.01

Country Specific Regression (M5 Specification)

Variable	Pooled (N=1,451)	India only (N=562)	Indonesia (N=889)
Post Treatment (H3)	−0.0298 (ns)	+0.0209 (ns)	−0.0301 (ns)
ΔSentiment (H1)	+0.0009 (ns)	+0.0000 (ns)	+0.0013 (ns)
R^2	0.02	0.013	0.028
Adj. R^2	−0.004	−0.029	−0.009

* p<0.1 ** p<0.05 *** p<0.01